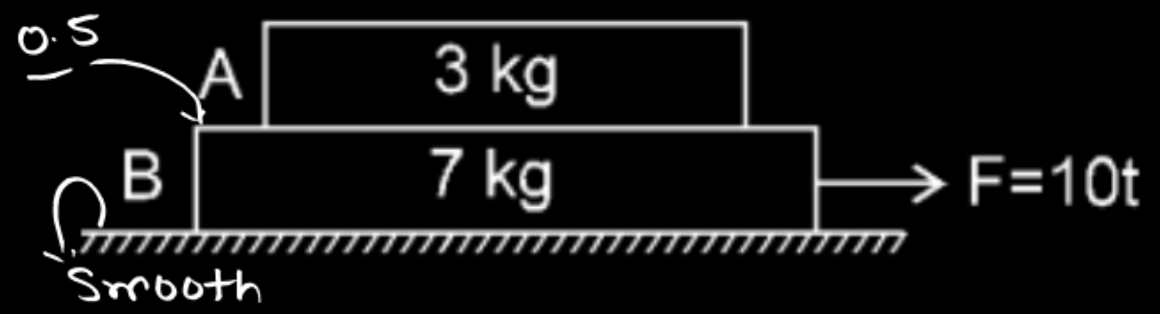
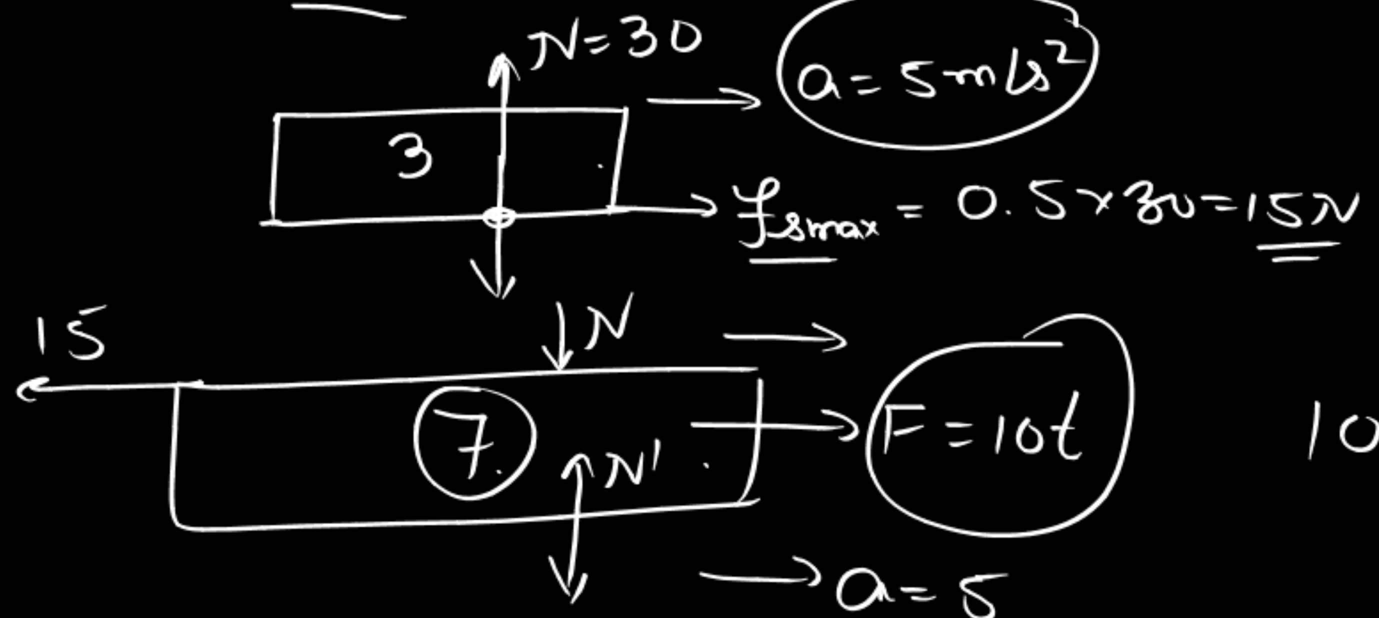
A variable force F = 10 t is applied to block B placed on a smooth surface. The coefficient of friction between A & B is 0.5. (t is time in seconds. Initial velocities are zero):



(A) block A starts sliding on B at t = 10 seconds (B) block A starts sliding on B at t = 6 seconds

(C) acceleration of A at 10 seconds is 10 m/s². (D) acceleration of A at 10 seconds is 5 m/s².

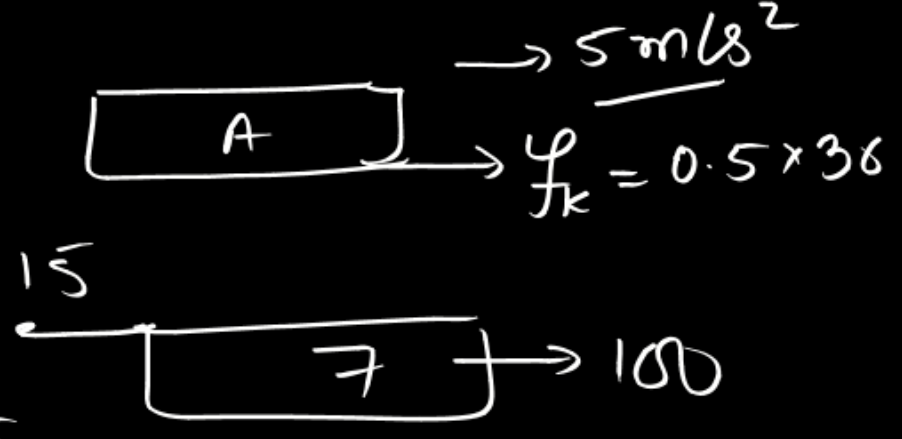
Sol :



$$10t - 15 = 7 \times 5$$

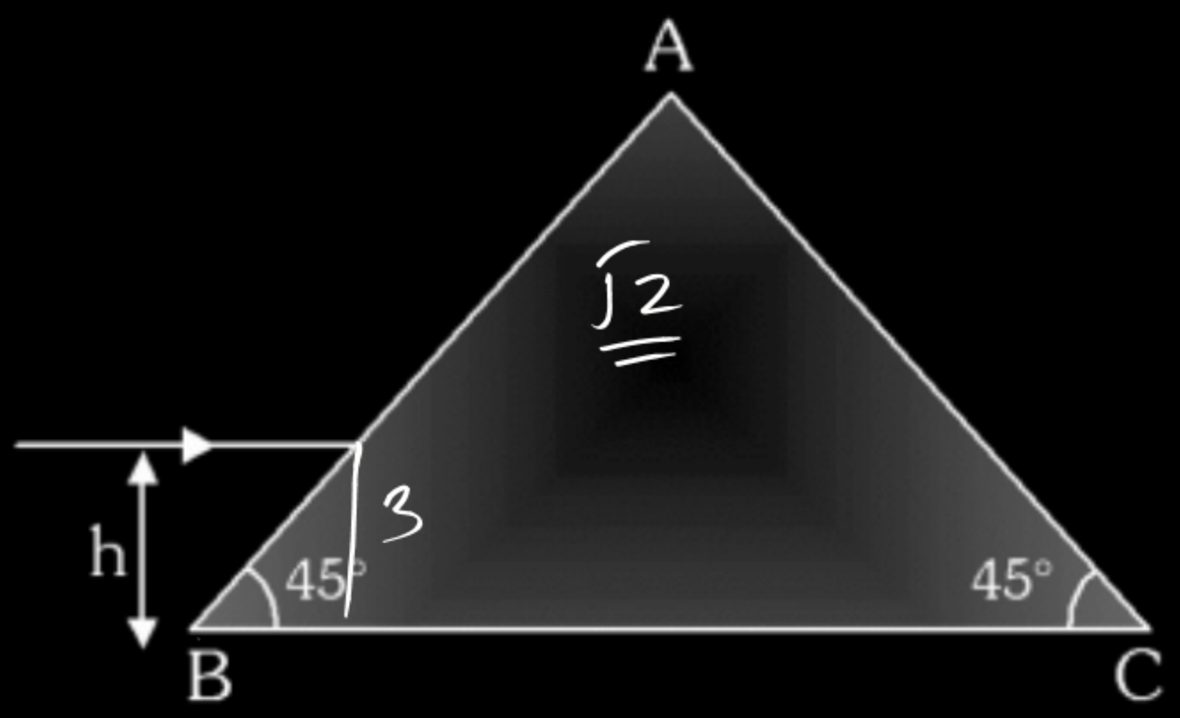
$$10t = 50$$

$$\boxed{t = 5\ sec}$$



$$\frac{85}{7} = a$$

A ray of light is incident parallel to BC at a height h = 3.0 cm from BC. Find the height (in cm) above BC at which the emergent ray leaves the surface AC. It is given that $\mu = \sqrt{2}$ and length BC = 20 cm. Take tan 15° = 0.25



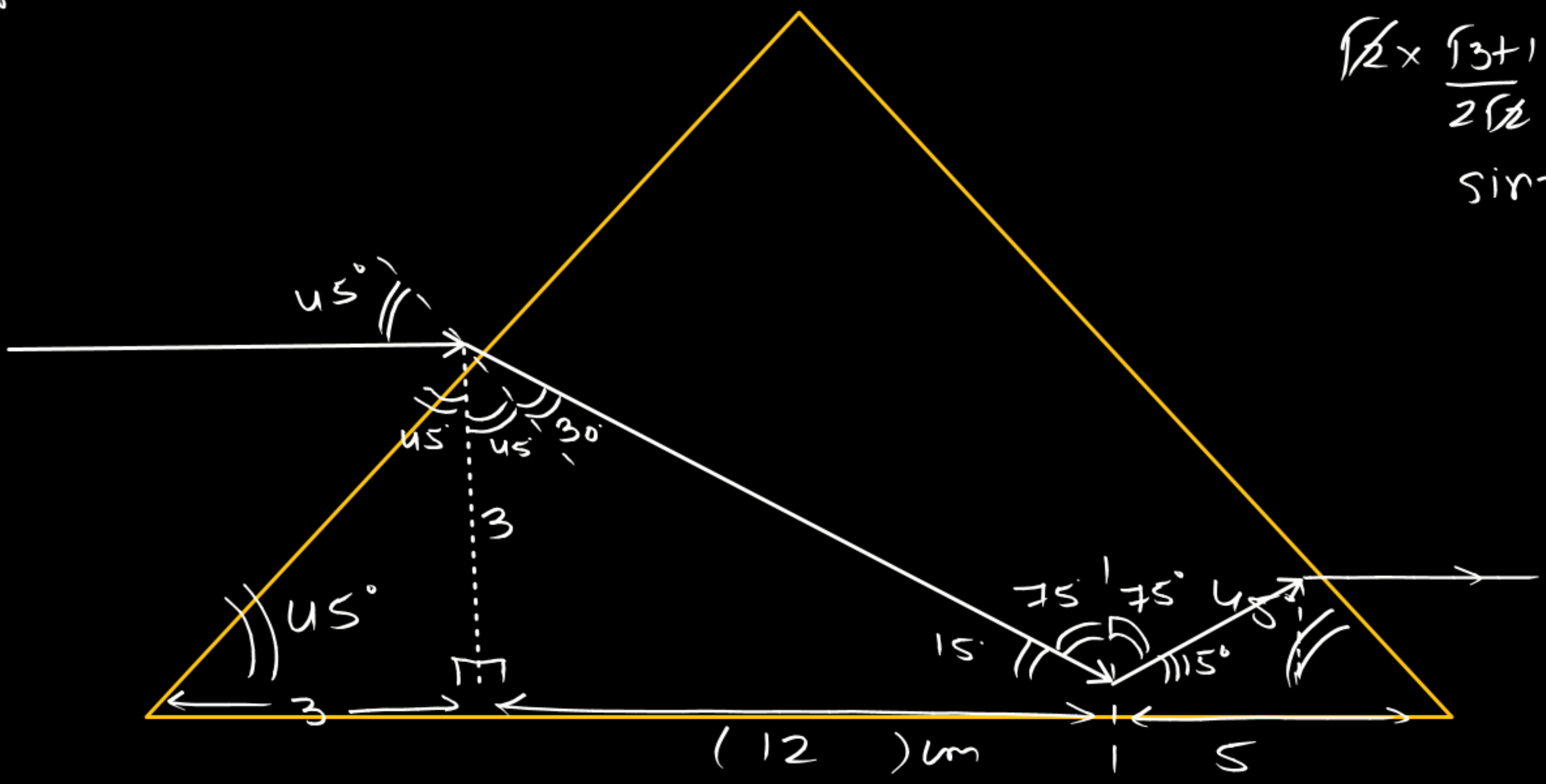
(A) 1 (B) 3 (C) 4 (D) 5

$1\times\frac{1}{\sqrt{2}}=\sqrt{2}\sin r$

$r=30^\circ$

$\sqrt{2}\times\frac{\sqrt{3}+1}{2\sqrt{2}}=1\times\sin r$

$\sin r=\frac{2.7}{2}>1$



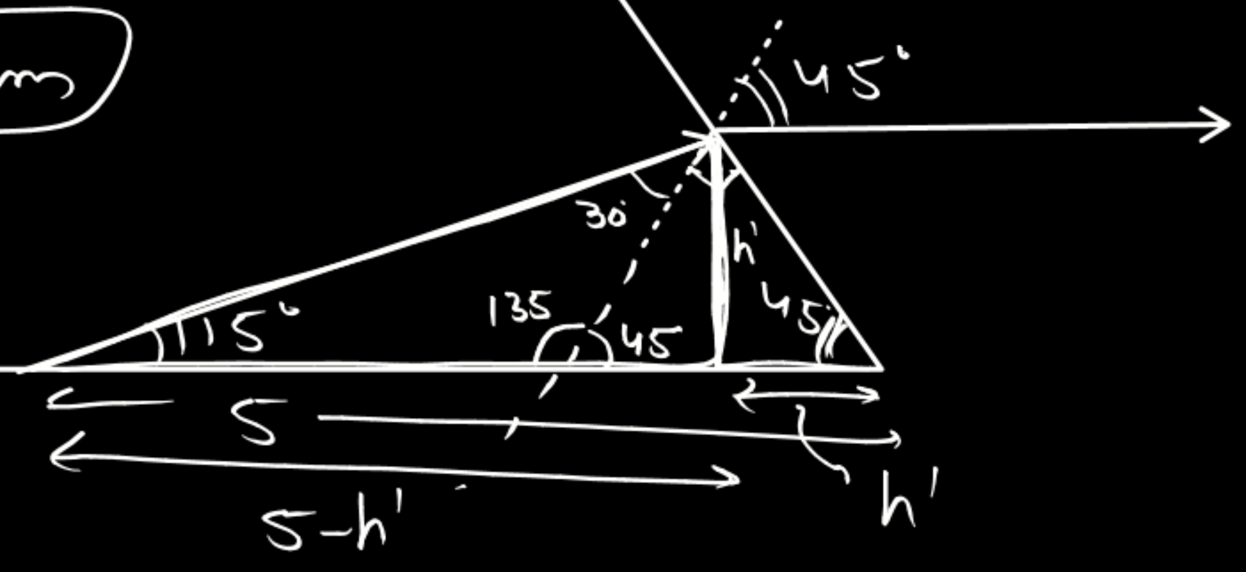
$\tan 15^\circ=\frac{3}{(\ \ )}=0.25$

$\boxed{(\ \ )=12}$

$$\tan 15^\circ = \frac{h'}{5-h'} = \frac{1}{4}$$

$$4h' = 5 - h'$$

$$\boxed{h' = 1\,\text{cm}}$$



Two radius stations broad cast their programmes at same amplitude A and at slightly different linear frequencies ω_1 and ω_2 respectively where $(\omega_2 - \omega_1) = 10^3$ Hz A detector receives the signals from two stations simultaneously, it can detect signals of intensity $\boxed{> 2A^2}$.

$f_{Beat} = 1000 Hz$

$T_{Beat} = \frac{1}{1000} = 10^{-3} sec$

Choose the correct statement(s) :

(A) The time interval between successive maxima of intensity of signal received by detector is 10^{-3} sec.

(B) The time interval between successive maxima of intensity of signal received by detector is 0.5×10^{-3} sec.

(C) The time for which the detector remains idle in each cycle of intensity of signal is 10^{-4} sec.

(D) The time for which the detector remains idle in each cycle of intensity of signal is 2×10^{-4} sec.

$I_{Res} = [0, 4A^2]$ $\quad I_1 = A^2 \quad I_2 = A^2$

$$I_{Res} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$$

$$= A^2 + A^2 + 2A^2\cos\Delta\phi = 2A^2\{1+\cos\Delta\phi\} = 4A^2\cos\frac{\Delta\phi}{2}$$

$$Y_1 = A \sin 2\pi\omega_1 t$$

$$Y_2 = A \sin 2\pi\omega_2 t$$

$$Y_{Res} = A\left\{\sin 2\pi\omega_1 t + \sin 2\pi\omega_2 t\right\}$$

$$= 2A \sin \pi(\omega_1+\omega_2)t \ \cos \pi(\omega_1-\omega_2)t$$

$$Y = \underbrace{\left\{2A \cos\pi(\omega_1-\omega_2)t\right\}}_{A'} \sin \pi(\omega_1+\omega_2)t$$

$$A' = 2A\cos\pi(\omega_1-\omega_2)t$$

$$I = (A')^2 = 4A^2 \cos^2\pi(\omega_1-\omega_2)t$$

$$\cos^2\pi(\omega_1-\omega_2)t = +1$$

$$\cos\pi(\omega_1-\omega_2)t = \pm 1$$

$$\pi \times 1000\, t = 0, \pi, 2\pi, \ldots$$

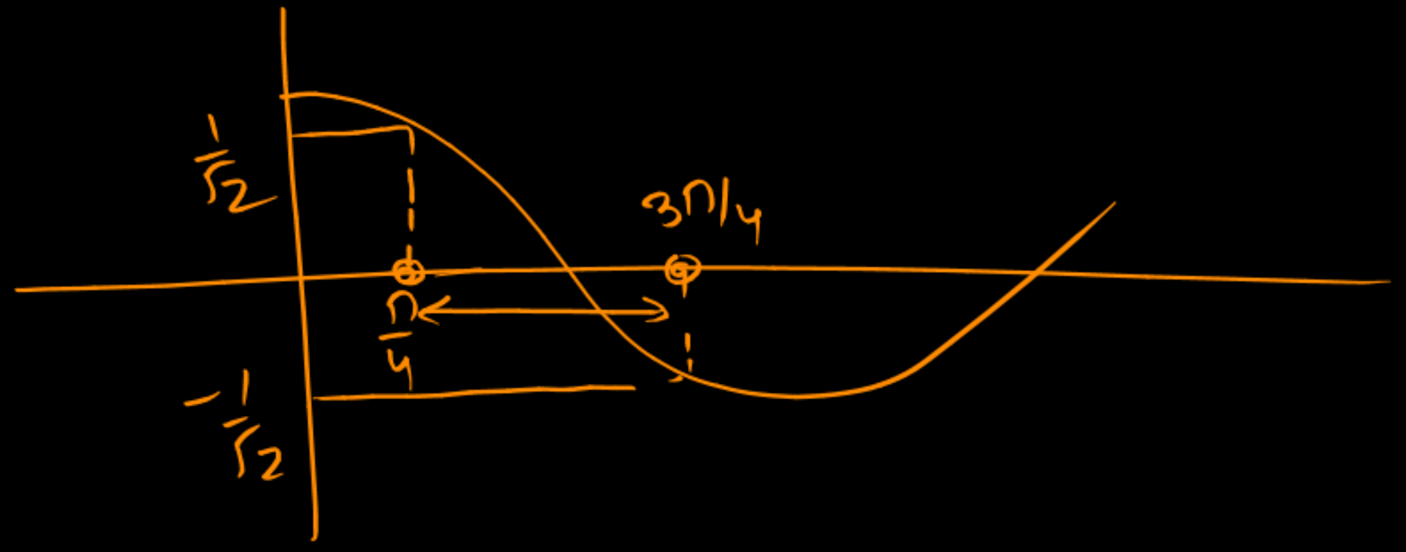
$$t = 0, \frac{1}{1000}, \frac{2}{1000}, \frac{3}{1000}$$

$$\frac{1}{1000} = 10^{-3}\,\text{sec}$$

$$I = 4A^2\cos^2\pi\Delta\omega t < 2A^2$$

$$\cos^2\pi\Delta\omega t < \frac{1}{2}$$

$$-\frac{1}{\sqrt{2}} < \cos\pi\Delta\omega t < \frac{1}{\sqrt{2}}$$



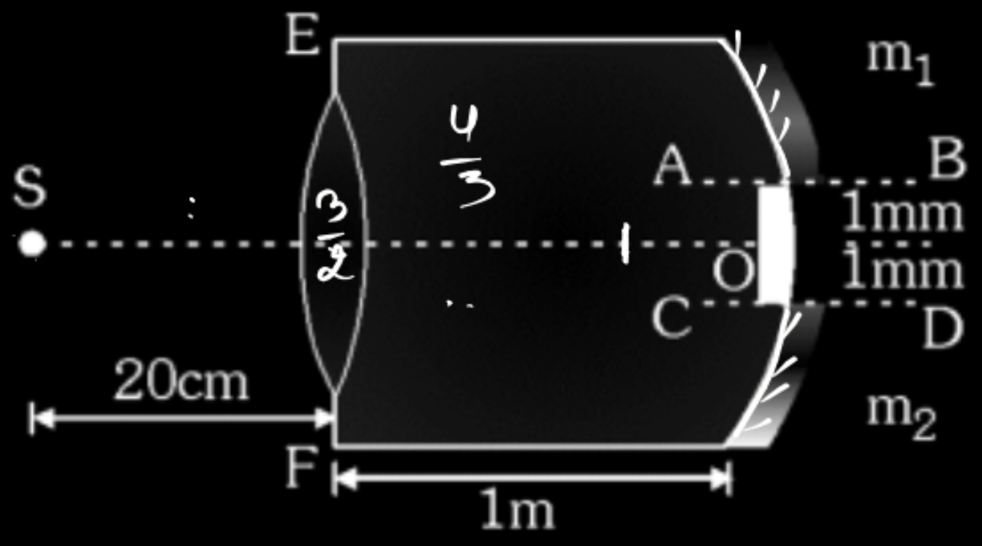
$$\pi \Delta\omega t_1 = \frac{\pi}{4}$$

$$\pi \Delta\omega t_2 = \frac{3\pi}{4}$$

$$\cancel{\pi} \Delta\omega (t_2 - t_1) = \frac{\cancel{\pi}}{2}$$

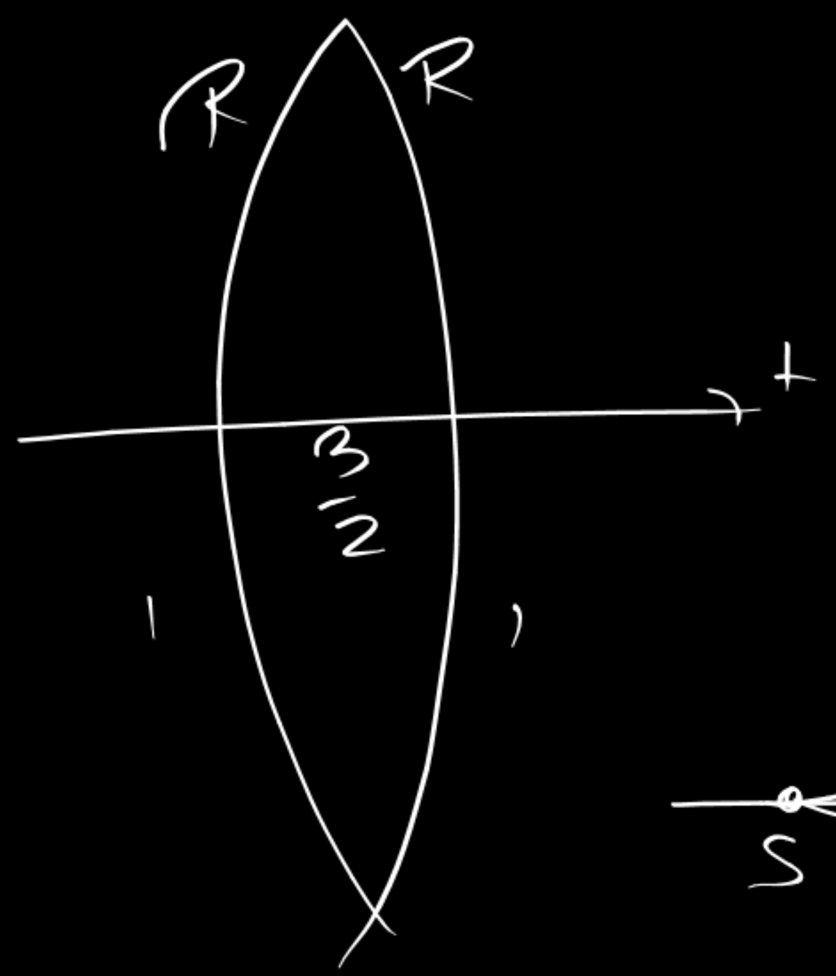
$$t_2 - t_1 = \frac{\cancel{\pi}}{2\times1000} = 0.5\times10^{-3}\,\text{sec}$$

An equiconvex lens of focal length 10 cm (in air) and refractive index 3/2 is put at a small opening on a tube of length lm fully filled with liquid of refractive index 4/3. A concave mirror of radius of curvature 20 cm is cut into two halves m_1 and m_2 and placed at the end of the tube. m_1 and m_2 are placed such that their principal axes AB and CD respectively are separated by 1 mm each from the principle axes of the lens. A slit S placed in air illuminates the lens with light of frequency 7.5×10^{14} Hz. The light reflected from m_1 and m_2 forms interference pattern on the left end EF of the tube. O is an opaque substance to cover the hole left by m_1 and m_2. Find :



The distance between the images formed by m_1 and m_2 is

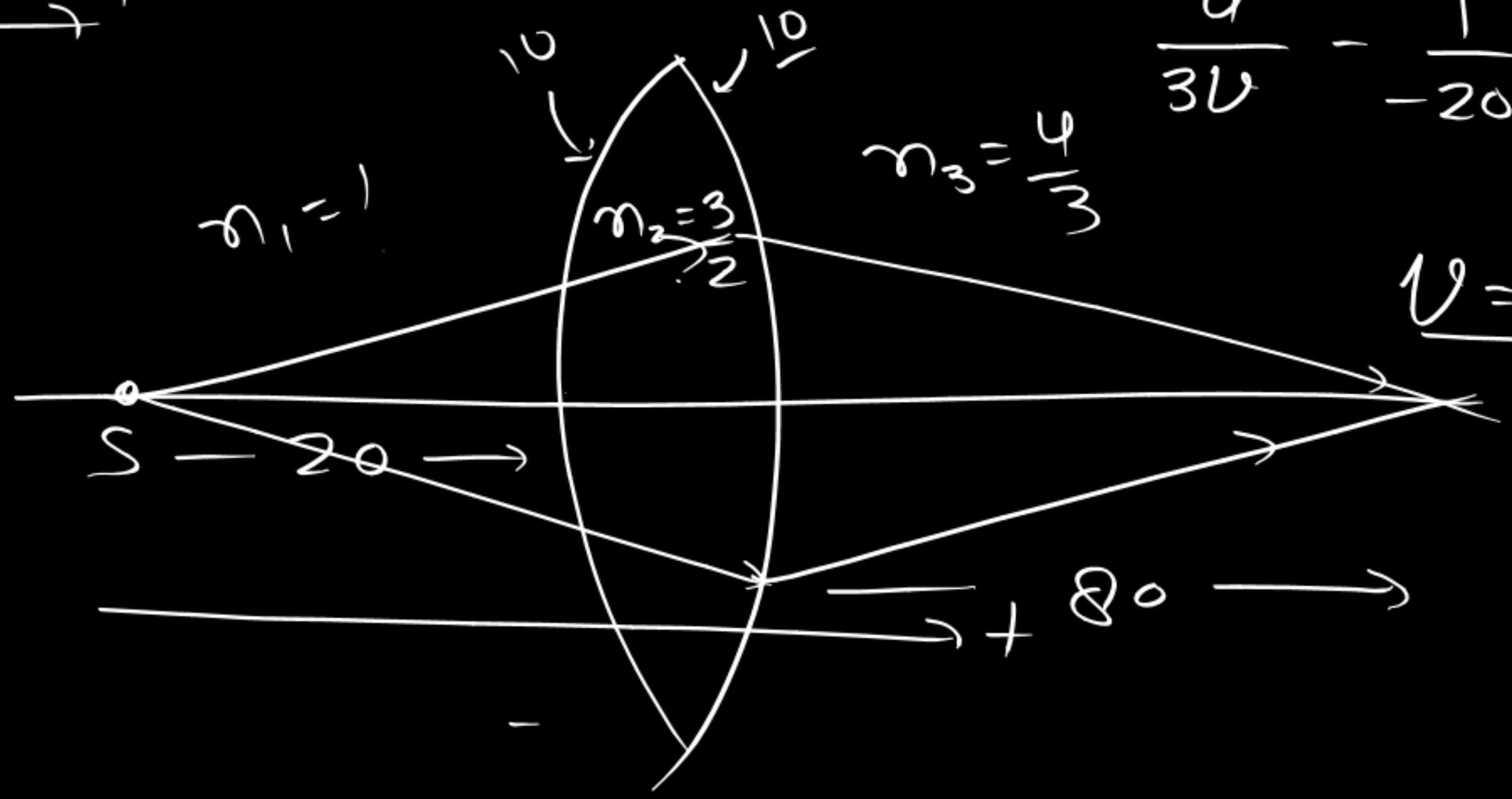
(A) 10 mm　(B) 4 mm　(C) 16 mm　(D) 24 mm



$f = 10\,cm$

$$\frac{1}{10} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{R} - \frac{1}{-R}\right)$$

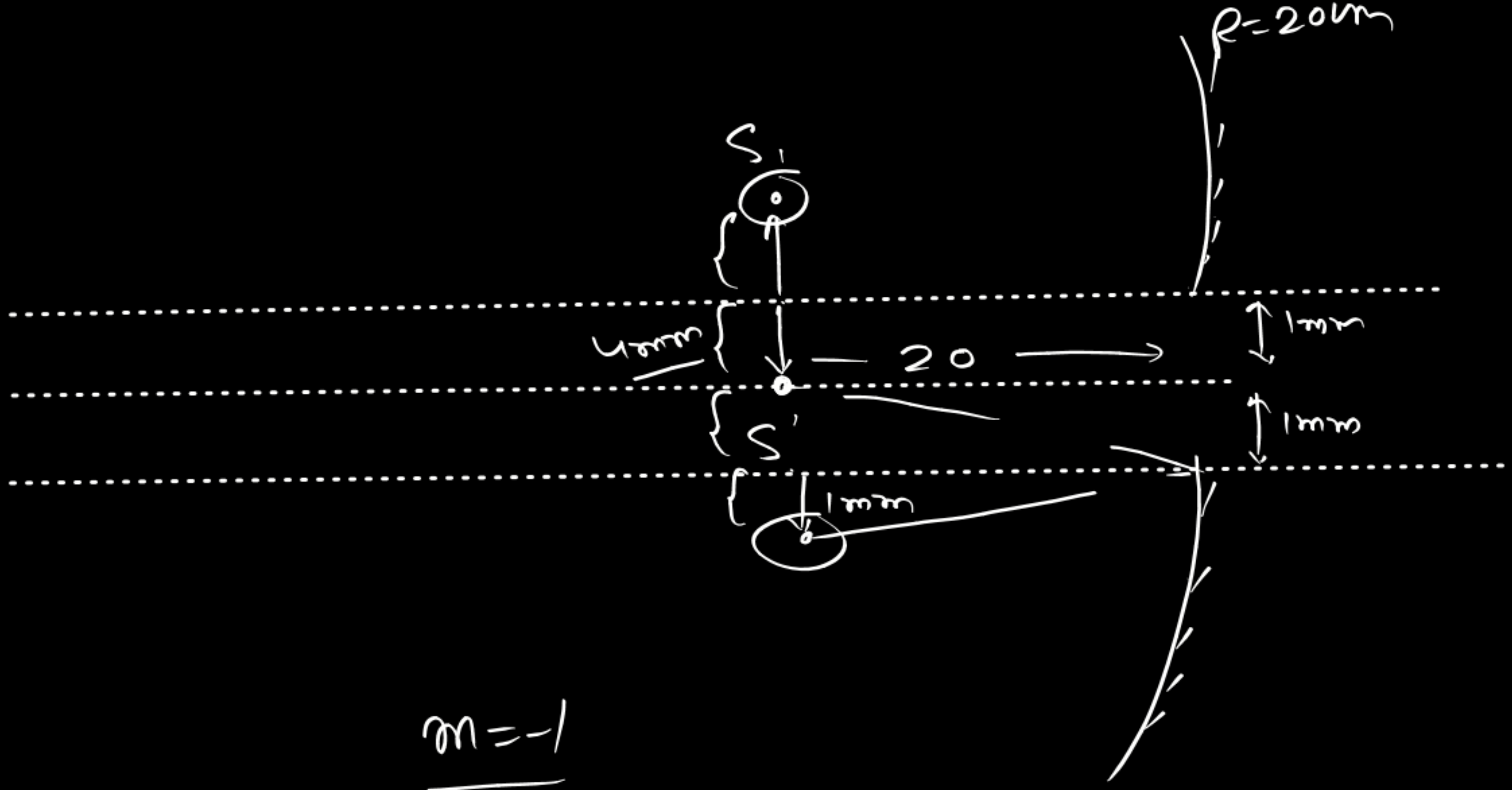
$$\boxed{R = 10\,cm}$$



$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}$$

$$\frac{4}{3v} - \frac{1}{-20} = \frac{1/2}{10} + \frac{\frac{4}{3} - \frac{3}{2}}{-10}$$

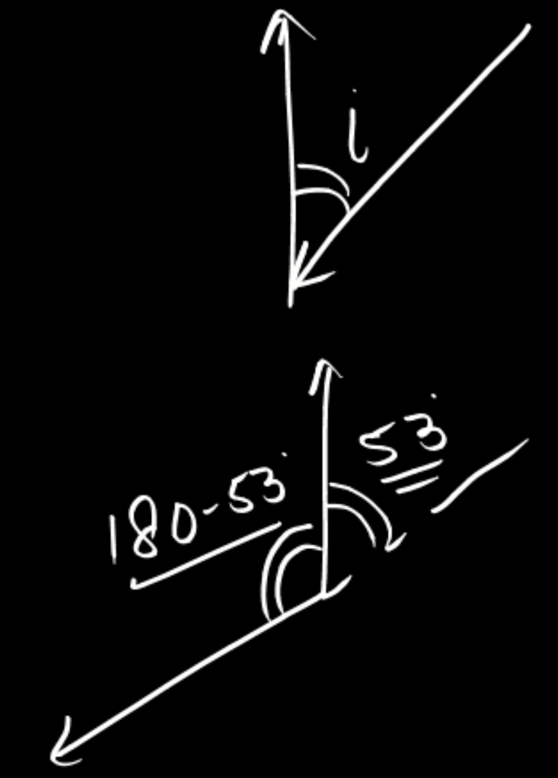
$$v = 80$$

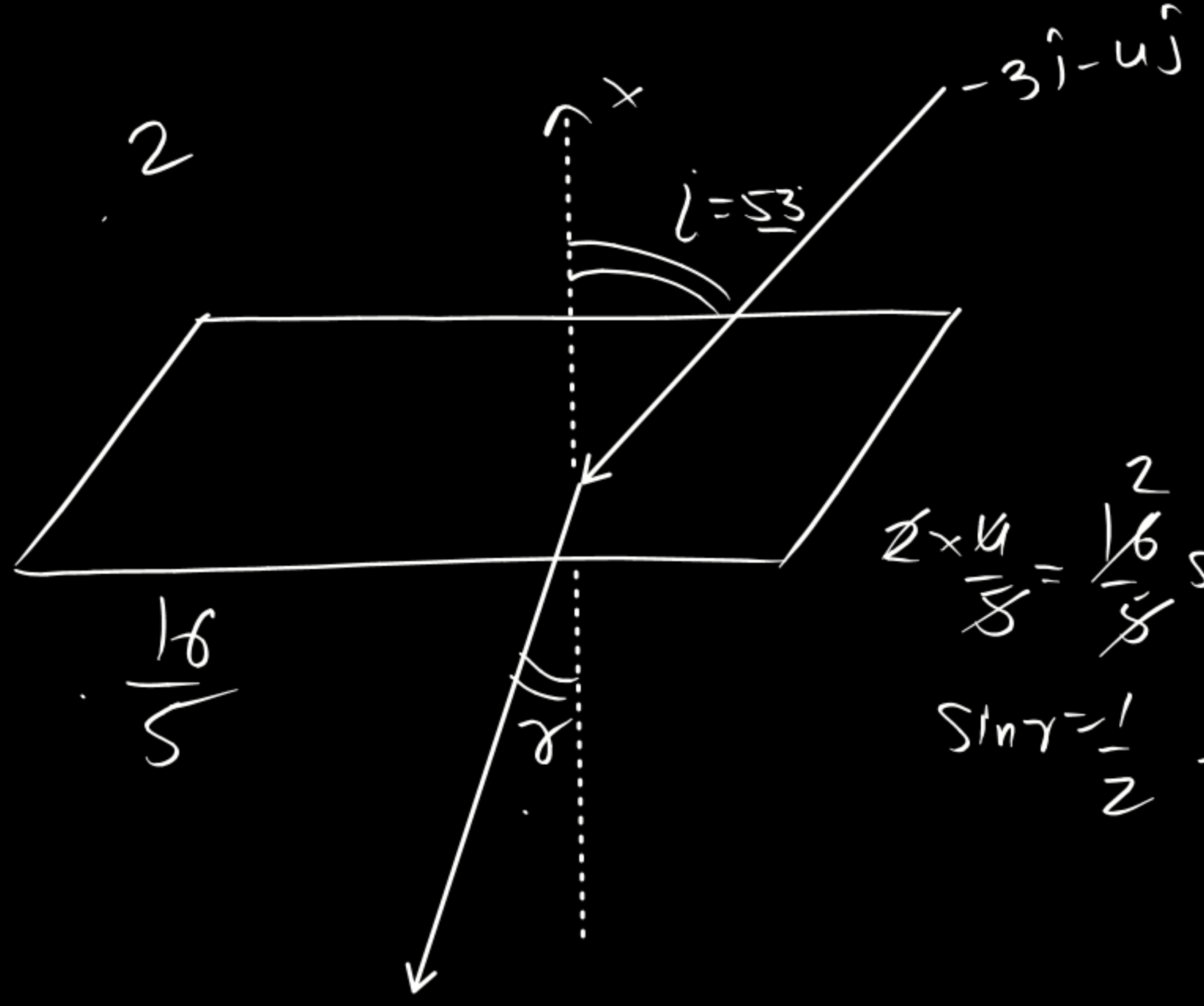


$m = -1$

A ray of light moving along $-3\hat{i}-4\hat{j}$ undergoes refraction at an interface of two media which is the y − z plane. The refractive index for $x > 0$ is 2 while for $x < 0$ is $\frac{16}{5}$. Find the angle of refraction if incident ray is in region $x > 0$

(A) ~~30º~~ (B) 45º (C) 37º (D) 60º





$\hat{x}$ $\hat{i}$ $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{ab}$

$\cos\theta = \frac{(-3\hat{i}-4\hat{j})\cdot\hat{i}}{5\times 1} = \frac{-3}{5}$ $\theta = 180^\circ - 53^\circ$

$\cancel{2}\times\frac{4}{\cancel{5}} = \frac{\cancel{16}}{\cancel{5}}\sin r$

$\sin r = \frac{1}{2}$ $r = 30^\circ$

A whistle of frequency f_0 =1300 Hz is dropped from a height M = 505 m above the ground. At same instant, a detector is projected upwards with velocity v = 50 m/s along the same line. If the velocity of sound is 300 m/s. Choose the correct statement(s) :

(A) The sound received by detector at t = 5 sec. is emitted by source at t = 3 sec. X

(B) The sound received by detector at t = 5 sec. is emitted by source at t = 2 sec. x

(C) The frequency of sound received by detector at t = 5 sec. is 1500 Hz (approximately).

(D) The frequency of sound received by detector at t = 5 sec. is 1800 Hz (approximately).

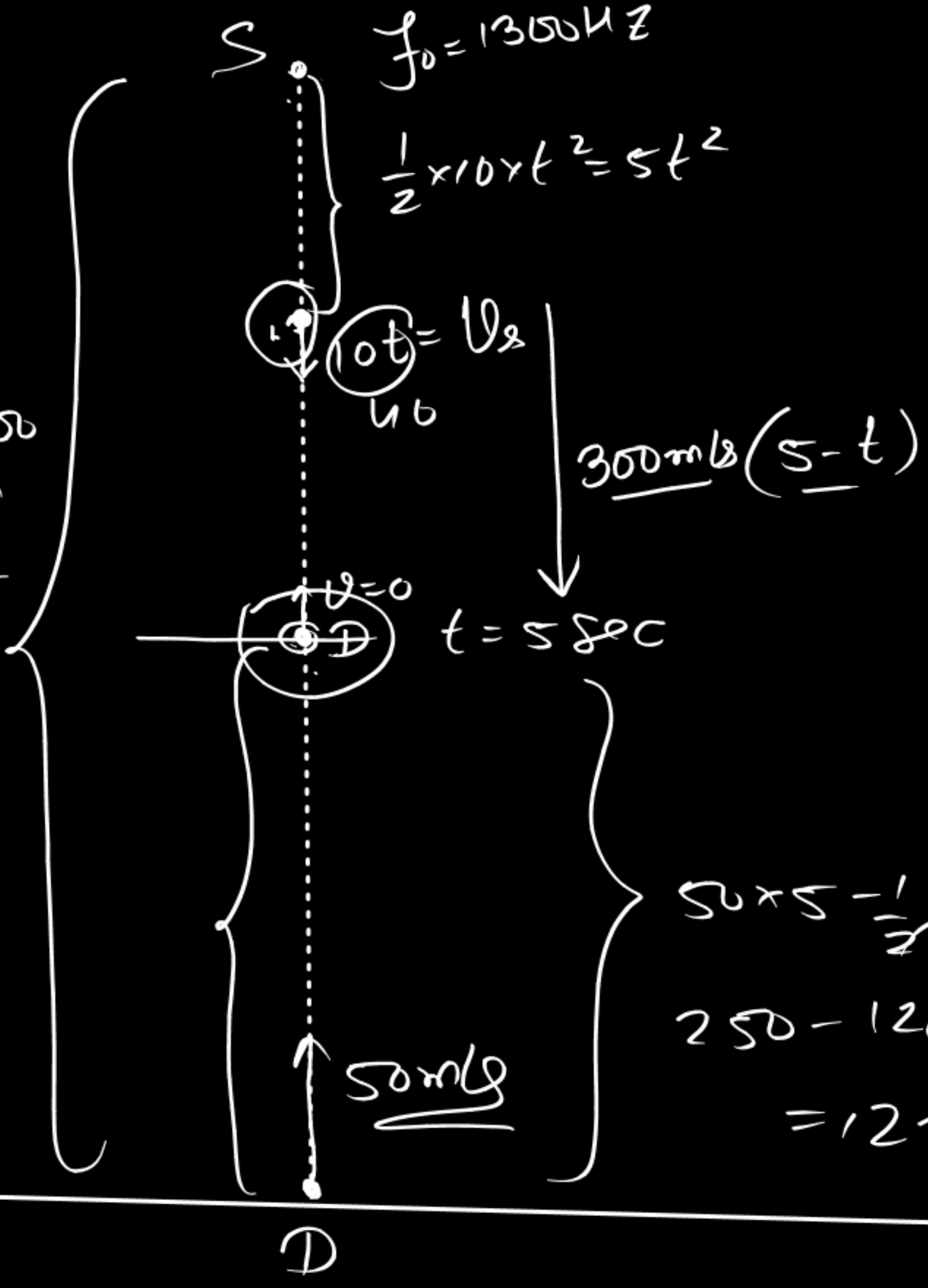
(i)

$$f' = \frac{c \pm v_0}{c \pm v_s} \times f_0$$

$$= \frac{300 + 0}{300 - 40} \times 1300$$

$$= \frac{300}{\cancel{260}} \times \cancel{1300}^{5}$$

$$= 1500\,\text{Hz}$$



$$50 \times 5 - \frac{1}{2} \times 10 \times 5^2$$

$$250 - 125$$

$$= 125$$

$$5t^2 + 300(5-t)$$

$$+ 125 = 505$$

$$5t^2 + 1500 - 300t + 125 = 505$$

$$5t^2 - 300t + 1500 - 380 = 0$$

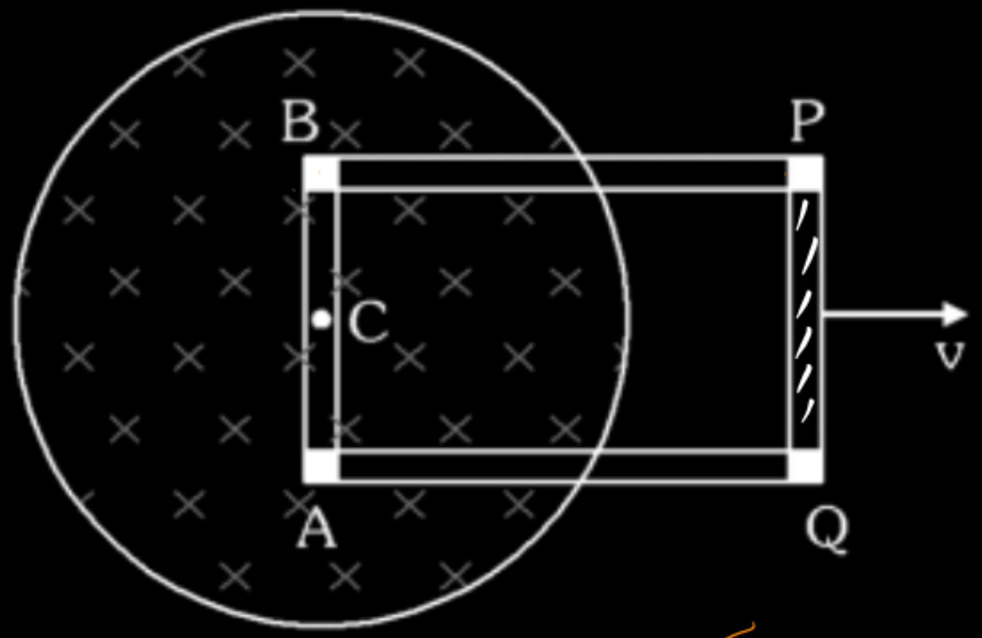
$$5t^2 - 300t + 1120 = 0$$

$$\boxed{t^2 - 60t + 224 = 0}$$

$$t = 4, \cancel{56}$$

$$\boxed{t = 4\,\text{sec}}$$

Uniform but time varying magnetic field $B = \frac{B_0 t}{T}$ present in the cylindrical region. There is small section at corners which is made with insulating material and other portion of rectangular loop is conducting. Loop is moving with constant velocity v. At t = T situation shown in figure. C is centre of cylindrical region. Choose correct option/s at given instant.

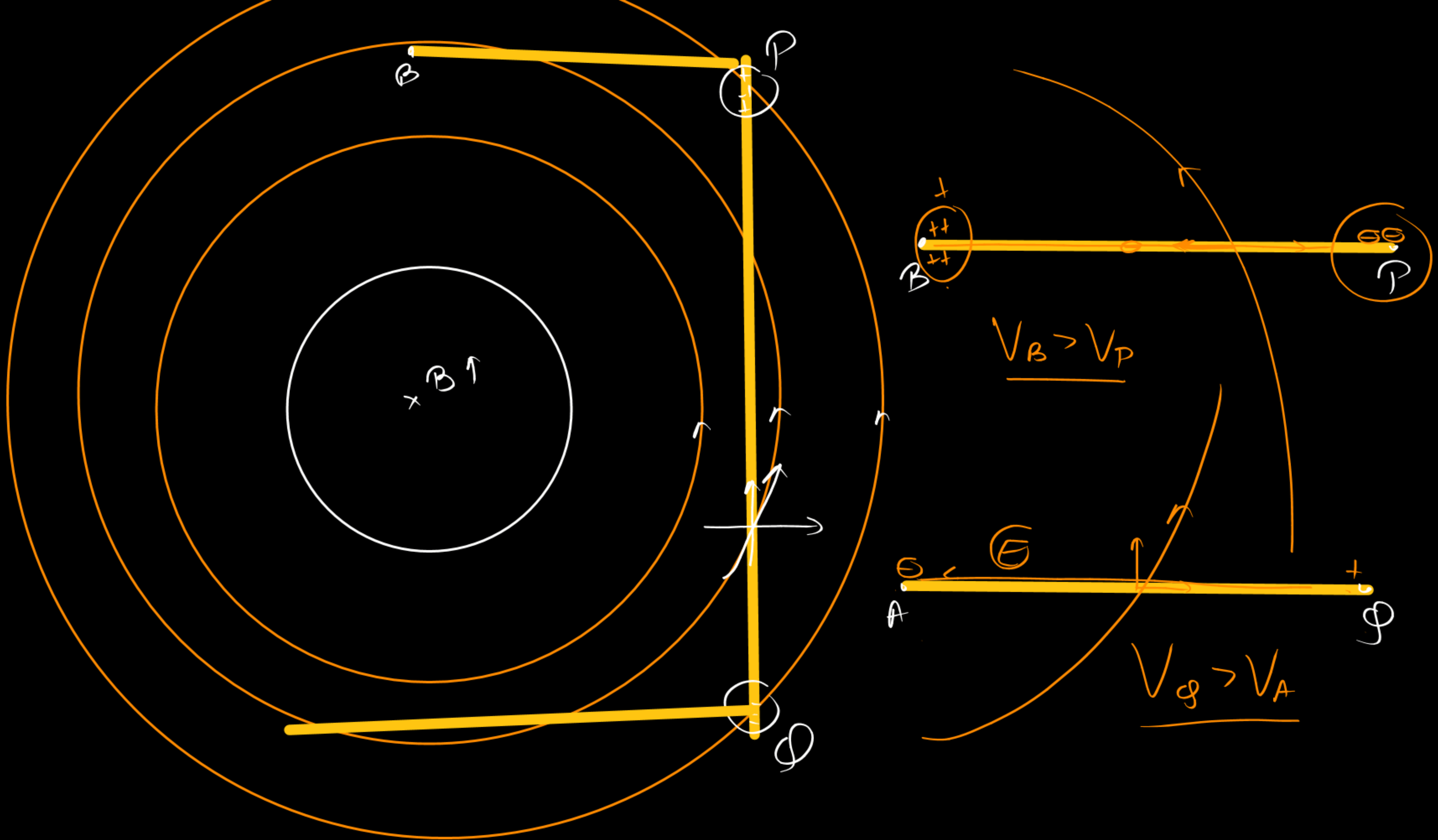


(A) There is no emf in section PQ

(B) From B to P potential decreases

(C) From A to Q potential decreases

(D) These is no electrostatics field in section PQ


B
P
B
Q
B
P
$V_B > V_P$
A
Q
$V_Q > V_A$

Two concentric metallic shells of radius R and 2R, out of which the inner shell is having charge Q and outer shell is uncharged. If they are connected with a conducting wire. Then,

(A) Q amount of charge will flow from inner to outer shell.

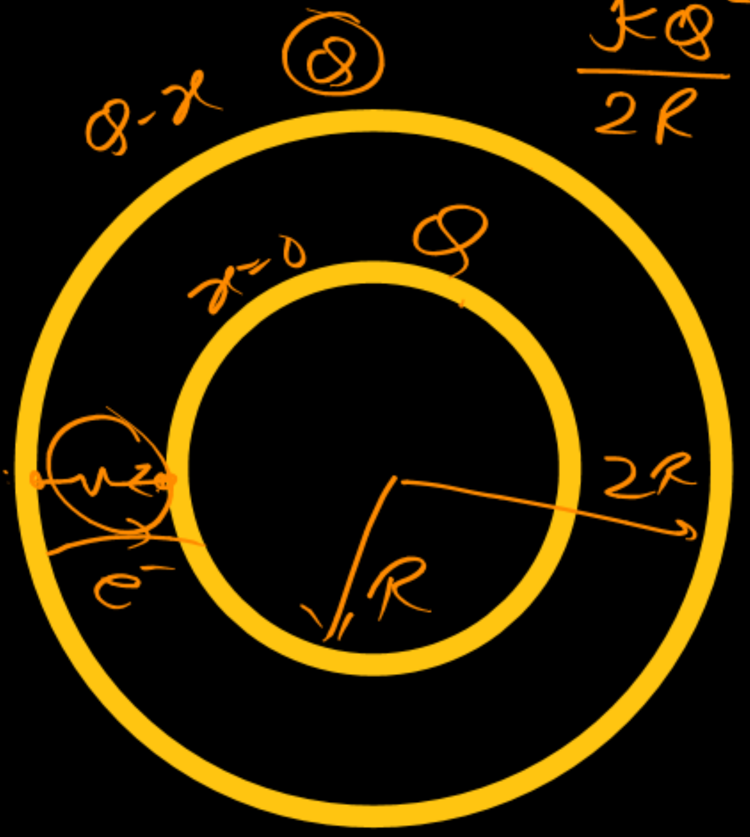
(B) Q/e number of electrons will flow from inner to outer shell, where e is charge of electron.

(C) $\frac{KQ^2}{8R}$ amount of heat is produced in the wire

(D) $\frac{KQ^2}{4R}$ amount of heat is produced in the wire.

$$\frac{K(Q-x)}{2R} + \frac{Kx}{2R} = \frac{Kx}{R} + \frac{K(Q-x)}{2R}$$

$$x = 0$$

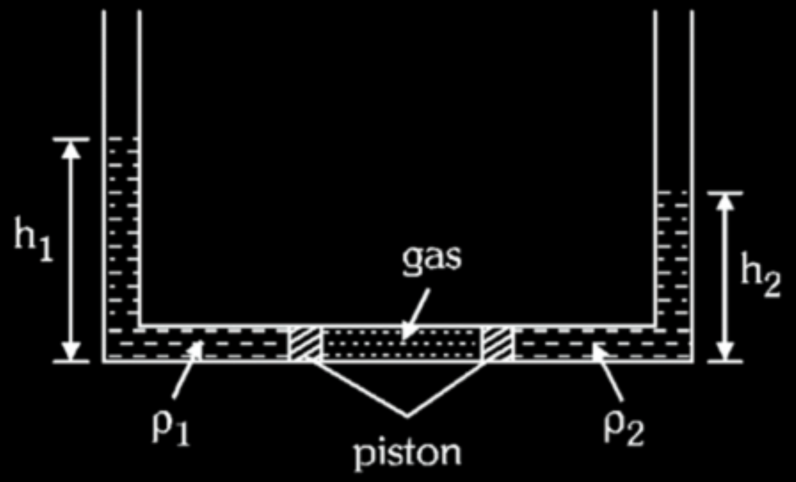


$U_i - U_f$

$$\frac{KQ^2}{2R} + 0 + 0 - \left\{\frac{KQ^2}{4R} + 0 + 0\right\}$$

$$\frac{KQ^2}{4R}$$

Figure shows a tube whose two limbs are vertical and one is horizontal of sufficient length. An ideal gas is enclosed between two massless non-conducting movable piston of cross-sectional area 10 cm^2. A liquid of density ρ_1= 1000 kg/m^3 and ρ_2 = 2000 kg/m^3 are poured in the vertical limbs of height of h_1 = 2m and h_2. Now gas is slowly heated. Consider the wall of tube is insulated.



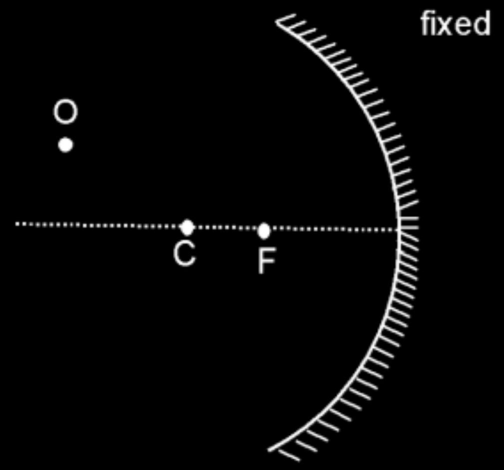
(A) Ratio of h_1 and h_2 at any instant will be 2/1

(B) If due to heating separation between piston is increased by 3m then displacement of piston on left will be lm

(C) If due to heating separation between piston is increased by 3m then displacement of piston on left will be 2m

(D) The work done by gas for above situation is 390 Joule

An object above principal axis is towards left of centre of curvature of mirror. Which of the following is/are correct at given instant (consider image formation by paraxial rays)



(A) If objects moves parallel to principal axis then image also must moves parallel to principal axis.

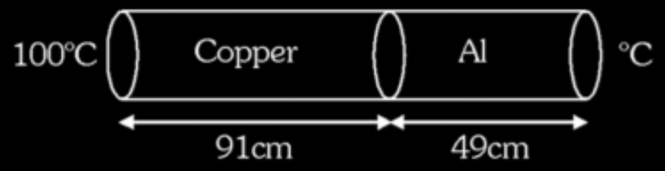
(B) If object moves perpendicular to principal axis then image also must moves perpendicular to principal axis.

(C) Object and image can move along same line.

(D) If object moves towards the mirror such that its line of motion is passing through focus then image will move parallel to principal axis.

A bar of Copper length 91 cm and Aluminium of length 49 cm are joined together end to end. Both rod havecircular cross-section with diameter $\sqrt{\frac{1}{50\pi}}\mathrm{m}$. Free end of Copper & Aluminium are maintain at 100°C and 0°Crespectively. For maintaining temperature steam at 100°C is continuously flowing at the end of copper & ice is continuously melting at the end of Aluminium. After reaching steady state, choose correct statement(s) [Thermalconductivity of copper is 91 cal $m^{-1}{}^{\circ}c^{-1}s^{-1}$and Thermal conductivity of Aluminium is 49 cal $m^{-1}{}^{\circ}c^{-1}s^{-1}$, L_f = 80 cal/g andL_v = 540 cal/g].

Neglect any thermal resistance at the junction.



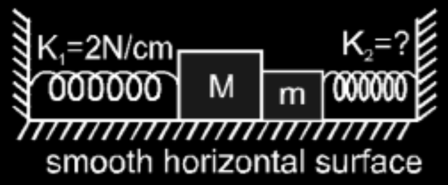
(A) Heat flow per unit time across the junction is 25 cal/sec.

(B) The rate of melting ice is $\frac{5}{16}\mathrm{g/sec}$.

(C) Rate of condensation of steam is $\frac{5}{108}\mathrm{g/sec}$.

(D) Temperature of junction is 50°C

Two blocks of mass M and m, are used to compress two different massless springs as shown. The left spring is compressed by 3 cm, while the right spring is compressed by an unknown amount. The system is at rest, and all surfaces fixed and smooth. Which of the following statements are true ?



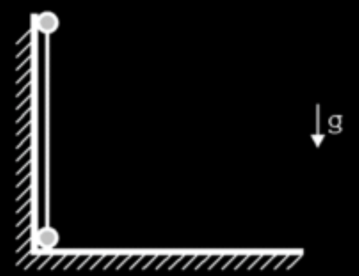
(A) The force exerted on block of mass m by the right spring is 6 N to the left.

(B) The force exerted on block of mass m by the right spring is impossible to determine.

(C) The net force on block of mass m is zero.

(D) The normal force exerted by block of mass m on block of mass M is 6 N.

A dumbbell constructed by affixing small identical balls each of mass m at the ends of a light rod of length ℓ stands vertically on a frictionless floor touching a frictionless wall as shown in the figure. If the lower ball is gently pushed away from the wall, the dumbbell begins to slide. Denoting acceleration due to gravity by g. Mark the **CORRECT** statements:



(A) The maximum speed of the lower ball is attained when upper ball looses contact.

(B) The maximum speed of the lower ball is attained when tension in the light rod is zero.

(C) The maximum speed of the lower ball is $\sqrt{\frac{8g\ell}{27}}$

(D) The speed of the upper ball when lower ball acquires maximum speed is $\sqrt{\frac{10g\ell}{27}}$

An isotropic point source of light of power P is kept in front of a screen which has a circular hole of radius R as shown in figure -1. Power crossing the circular hole is P1. Now a transparent cylinder which does not absorb any light of a refractive index $\mu(=\sqrt{2})$ is introduced between the source and screen and kept as shown in figure -2. Consider the following two cases.

Case-1 : Source is kept on the axis of the cylinder, just outside the cylinder.

Case-2 : Source is kept on the axis of the cylinder, just inside the cylinder.

In case-1 power crossing the circular hole is P2 and in case-2 it is P_3.

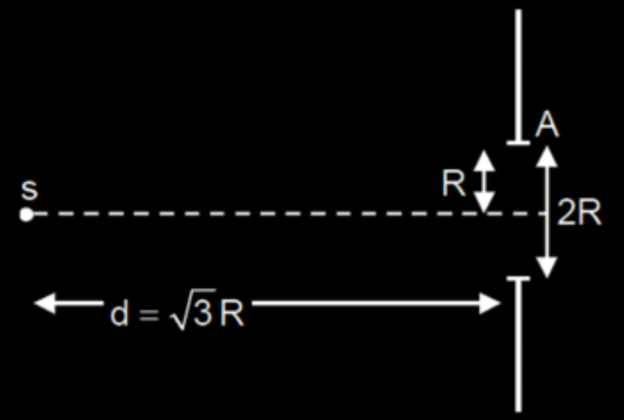


Figure -1

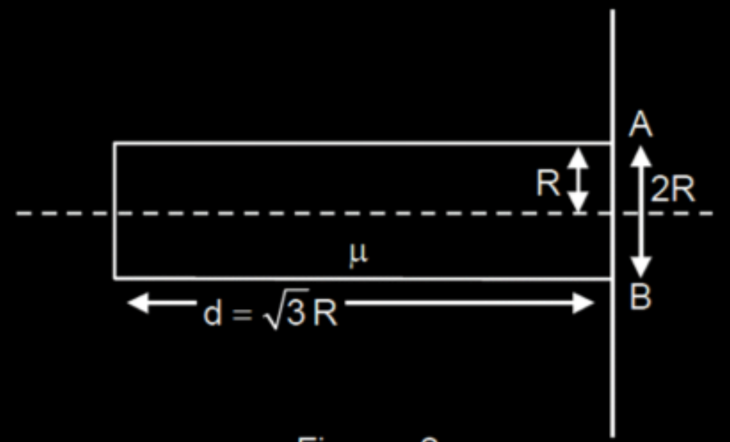


Figure -2

Answer the following two questions :

Value of P_2 is :

(A) $\frac{2P_1}{2-\sqrt{3}}$ (B) $\frac{P_1}{2-\sqrt{3}}$ (C) $\frac{2P_1}{2+\sqrt{3}}$ (D) $\frac{P_1}{2+\sqrt{3}}$

An isotropic point source of light of power P is kept in front of a screen which has a circular hole of radius R as shown in figure -1. Power crossing the circular hole is P1. Now a transparent cylinder which does not absorb any light of a refractive index $\mu(=\sqrt{2})$ is introduced between the source and screen and kept as shown in figure -2. Consider the following two cases.

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Case-2 : Source is kept on the axis of the cylinder, just inside the cylinder.

In case-1 power crossing the circular hole is P2 and in case-2 it is P_3.

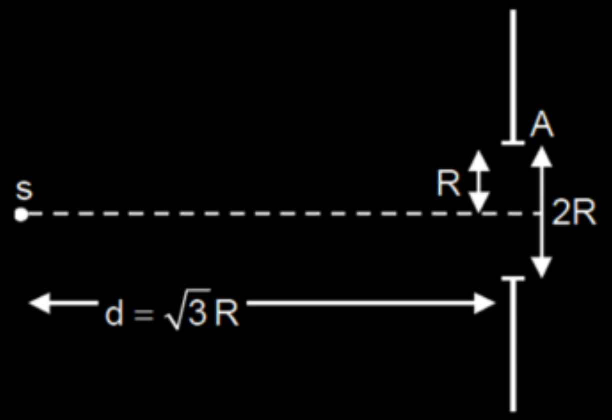


Figure -1

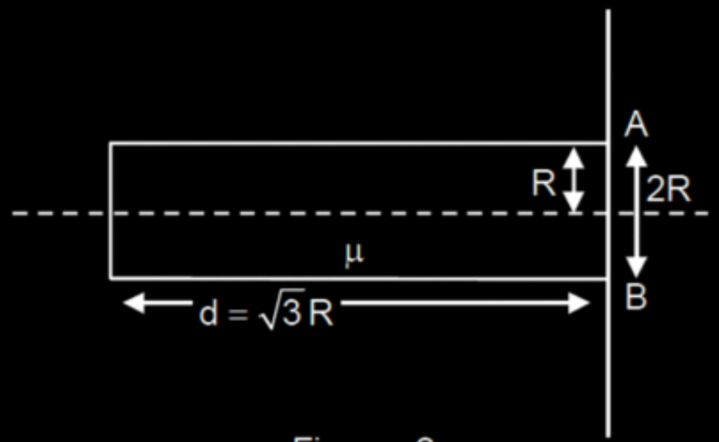


Figure -2

Answer the following two questions :

Value of P3 is :

(A) $P_1\left(\frac{2+\sqrt{2}}{2+\sqrt{3}}\right)$ (B) $P_1\left(\frac{2+\sqrt{2}}{2-\sqrt{3}}\right)$ (C) $P_1\left(\frac{2-\sqrt{2}}{2+\sqrt{3}}\right)$ (D) $P_1\left(\frac{2-\sqrt{2}}{2-\sqrt{3}}\right)$

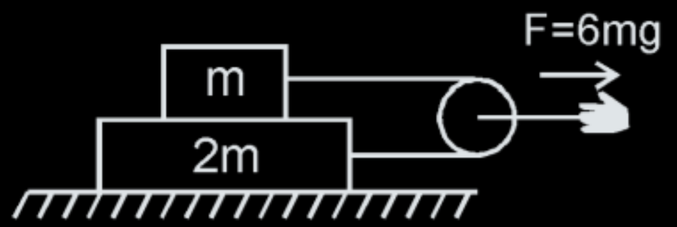
A radio consists of two isotopes. One of the isotopes decays by α-emission and the other by β-emission with half life T_1 = 405 second and T_2 = 1620 second, respectively, At t = 0, probabilities of getting α and β-particles from the radionuclide are equal.

If at t = 0, total number of nuclei in the ratio nuclide are N_0, calculate time t/100 (in sec.) when total number of nuclei remained undecayed becomes equal to $\frac{N_0}{2}$.

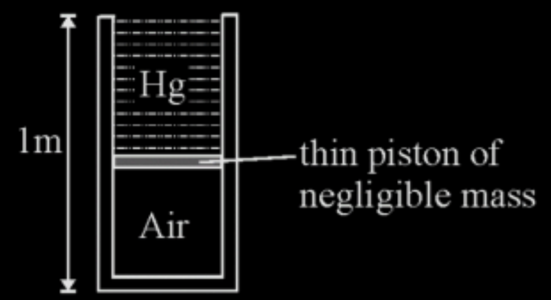
Given $\log_{10} 2 = 0.30103$ $\log_{10} 5.94 = 0.7742275$; $X^4 + 4x - 2.5 = 0 \Rightarrow x = 0.594$

A solid sphere of diameter 0.1 m is at 427°C and is kept in an enclosure at 27°C. Take Stefan's constant $= \frac{20}{3} \times 10^{-8}$ W/m^{2}K^4, emissivity of the surface 0.84, specific heat 0.1 kcal/kg K, density = 9280 kg/m^3, J = 4200 J/k cal. If rate of decrease of temperature of the sphere is N × 10^{-3}°C/s, find $\frac{N}{40}$.

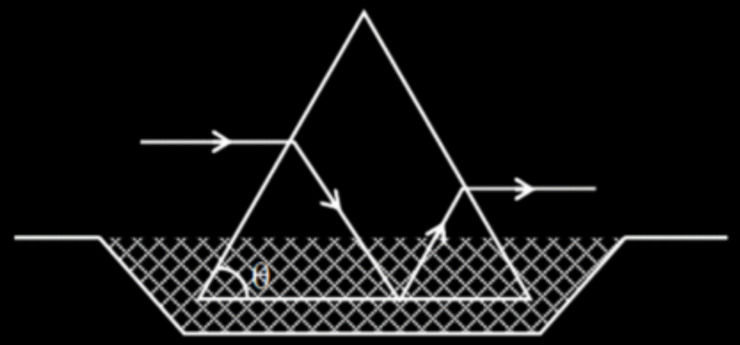
A block of mass m is placed on top of a block of mass 2m which in turn is placed on fixed horizontal surface. The coefficient of friction between all surfaces is $\mu = 1$. A massless string is connected to each mass and wraps halfway around a massless and frictionless pulley, as shown. The pulley is pulled by horizontal force of magnitude F = 6 mg towards right as shown. If the magnitude of acceleration of pulley is $\frac{X}{2}$ m/s^2, fill the value of X. (Take $g = 10\ m/s^2$)



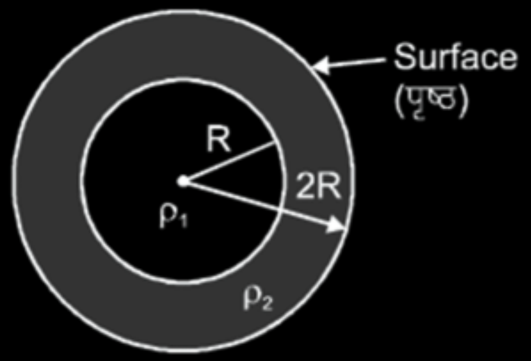
The 1 m tall conducting cylinder in figure contains air at a pressure of 76 cm of Hg. A very thin piston of negligible mass is placed at the top of the cylinder to prevent any air from air escaping, then mercury is very slowly poured into the cylinder until no more can be added without the cylinder over flowing. What is the height h (in cm) of the compressed air? Fill h/19 in OMR



An isosceles triangular glass prism stands with its base in water as shown. A ray is incident parallel to base of prism. After refraction through inclined face it just suffers total internal reflection on water glass interface. If value of $\tan\theta$ is $\frac{\lambda}{\sqrt{17}}$ then find λ. (Take $\mu_{glass} = 3/2$, $\mu_{water} = 4/3$)



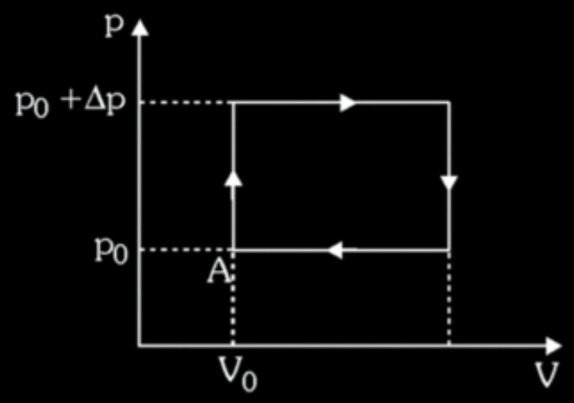
A planet is made of two materials of density 'ρ_1' and 'ρ_2' as shown in figure.



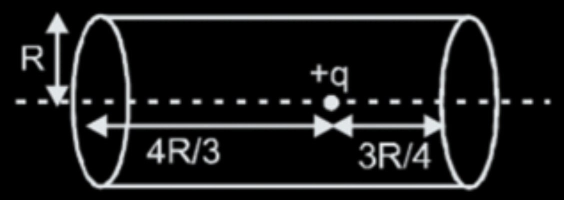
Acceleration due to gravity at surface of planet is same as that at depth 'R'. The ratio $\frac{\rho_1}{\rho_2}$ is equal to $\frac{x}{3}$.

Find the value of 'x'.

The working substance of a heat engine is a certain amount of monoatomic ideal gas. The gas is taken through the cyclic process shown in the figure which consists of two isobaric and two isochoric processes. Net work done by gas in a cycle is same as the internal energy of the gas in its initial state A. If for maximum possible efficiency of the cycle $\Delta p = np_0$, then find the value of $2n^2$.



A point charge +q is placed on the axis of a closed cylinder of radius R and length $\frac{25R}{12}$ as shown. If electric flux coming out from the curved surface of cylinder is $\frac{xq}{10 \epsilon_0}$, then calculate x.



An ideal DC source of emf E is connected with an uncharged capacitor and inductor. If switch is closed at $t = 0$, what is the maximum charge (in μC) on the capacitor in subsequent flow of charge ?

(Given : $L = 4H$; $C = 4.7\ \mu F$ and $E = 5.3$ volt)